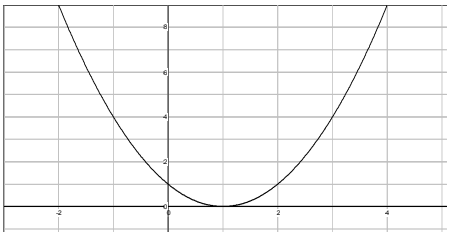
Question	Answer	Marks	Partial Marks
5(a)	[0]38 or [0]37.9 or [0]37.87...	**2**	**M1** for tan = $\frac{350}{450}$ oe If 0 scored, **SC1** for answer [0]52 or [0]52.1 or [0]52.12 to [0]52.13
5(b)	624 or 623.8 to 623.9	**6**	**M2** for 450 – 400 sin 50 or **M1** for sin 50 = $\frac{...}{400}$ **M2** for 350 + 400 cos 50 or **M1** for cos 50 = $\frac{...}{400}$ **M1** for (*their* (450 – 400 sin 50))^2 + (*their* (350 + 400 cos 50))^2
5(c)	10 min 8 s	**4**	**B3** for 10.1 or 10.13… or **M2** for (400 + 350 + 450 + *their DA*) ÷ 3 [÷ 60] oe or **M1** for any distance ÷ 3 **M1** for rounding *their* minutes into minutes and seconds to nearest second if clearly seen
6(a)	256	**1**	
6(b)	8	**2**	**M1** for $3(x^2 + 1) + 2$ or for 3(2) + 2
6(c)	$9x^2 + 12x + 5$	**3**	**M1** for $(3x + 2)^2 + 1$ **B1** for [$(3x + 2)^2$ =] $9x^2 + 6x + 6x + 4$ oe
6(d)	16	**2**	**M1** for $3x + 2 = 7^2 + 1$ or better
6(e)	$\frac{x-2}{3}$ oe final answer	**2**	**M1** for $x = 3y + 2$ or for $y - 2 = 3x$ or for $\frac{y}{3} = x + \frac{2}{3}$
6(f)	$\frac{4x^2 + 2x + 1}{3x + 2}$ final answer	**3**	**B1** for $x^2 + 1 + x(3x + 2)$ or better seen **M1** for common denominator $3x + 2$
6(g)	16	**1**	
7(a)	0.1	1	
7(b)(i)	0.2 oe 0.6, 0.3, 0.1 oe	**2**	**B1** for 0.2 **B1** for 0.6, 0.3, 0.1
7(b)(ii)	0.48 oe	**2**	**FT** *their* 0.6 from tree diagram **M1** for 0.8 × *their* 0.6

Question	Answer	Marks	Partial Marks
7(b)(iii)	0.28 oe	**3**	**M2** for $0.2 + 0.8 \times 0.1$ oe or **M1** for 0.2 or 0.8×0.1 or $0.8 \times (0.6 + 0.3)$
7(c)	0.32 oe	**3**	**M2** for $0.8 \times 0.2 + 0.2 \times 0.8$ oe **M1** for one of these products
8(a)(i)	36	**2**	**M1** for $\left(\frac{8}{12}\right)^2$ or $\left(\frac{12}{8}\right)^2$ oe
8(a)(ii)	30	**3**	**M2** for $320 \div 16 \times \frac{12}{8}$ oe or **M1** for $320 \div 16$
8(b)	3.375 cao	**3**	**M2** for $\frac{\frac{4}{3}\pi \times 4.5^3}{\pi \times 6^2}$ or better or **M1** for $\pi \times 6^2 \times h = \frac{4}{3} \times \pi \times 4.5^3$
8(c)	3.63 or 3.627 to 3.628	**3**	**M2** for $\frac{20^3}{40 \times \frac{4}{3}\pi}$ or **M1** for $40 \times \frac{4}{3} \times \pi \times r^3 = 20^3$
8(d)	$\frac{3x}{2}$ or $1.5x$ or $1\frac{1}{2}x$	**3**	**B2** for $4R^2 = 9x^2$ oe or better or **M1** for $4\pi R^2 = 2\pi x^2 + \pi \times 2x \times \frac{7x}{2}$
9(a)(i)	$(x+4)^2 - 25$	**2**	**B1** for $(x+k)^2 - 9 - (their\,k)^2$ or $(x+4)^2 - h$ or $k = 4$
9(a)(ii)	$x + 4 = [\pm]\ 5$	**M1**	**FT** *their* **(a)(i)**
	–9 and 1	**A1**	
9(b)	[b =] 7 [c =] –3	**3**	**B1** for [b =] 7 **M1** for $b^2 - 4c = 61$
9(c)(i)(a)	Correct sketch 	**2**	**B2** for correct quadratic curve with min touching x-axis or **B1** for parabola vertex downwards